

ZigBee® RF Component Solutions from RFMD

Solutions

RFMD has an extensive portfolio of single-chip integrated front end modules (FEMs) developed for high performance AMI and ZigBee applications in the 2.4 GHz to 2.5 GHz ISM bands. RF5745 and the RF65xx series are specifically designed to address the need for aggressive size reduction in a typical IEEE 802.15.4 front end design, offering reduced footprint and minimized component count outside the core chipset. Each of the RFMD FEMs meets or exceeds the RF front end needs for 802.15.4 RF systems and are fully DC- and RF-tested.

RFMD FEMs have architectures consisting of combinations of power amplifiers with:

- Tx harmonic filters
- Rx path linear amplifiers
- Antenna switches
- Switches used to combine the Rx and Tx paths at the transceiver interfaces

The RF65xx/RFFM620x series of modules address multiple RF transceiver interface configurations:

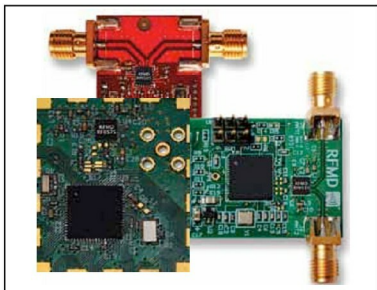
- Single port bi-directional
- Single-ended and differential
- Dual-port separate Rx and Tx single-ended
- Differential with integrated baluns

Logical interfaces have been developed that match transceiver requirements for minimal current draw and standard voltage levels. The RF antenna ports are matched to 50Ω and incorporate integrated DC-blocking capacitors.

Joint Reference Designs for ZigBee Solutions

Working jointly with major ZigBee chipset providers to produce reference designs allows RFMD to more easily support product development. It also enables customers to shorten time-to-market, cut costs, and reduce risks. Joint reference designs are themselves proof of platform cost, size, and performance.

These reference designs address manufacturers' needs for proven design technology minimizing power consumption and providing range extension at higher output levels from 100 mW to 500 mW. The reference designs are fully tested, with complete documentation sets available consisting of schematics, bills of material, and design files in standard Gerber format located on our website at www.rfmd.com/ZRG0611.



SPECIFICATIONS

Part Number	Architecture	Freq Range (MHz)	Gain (dB)	Avg Pout (dBm)	Vcc (V)	Current at Po (mA)	Package (dim. in mm)
RF6555	PA, SP2T, DP2T, LNA	2400 to 2500	25.0	18.0	2.0 to 3.6	70	5.0 x 5.0 x 1.0
RF6515	PA, SP2T	2400 to 2500	28.0	20.0	3.3	160	3.5 x 3.5 x 0.5
RF6525	PA, DP2T, LNA	2400 to 2500	28.0	20.0	3.3	160	3.5 x 3.5 x 0.5
RF6545	PA, DP2T, SPDT	2400 to 2500	28.0	20.0	3.3	160	3.5 x 3.5 x 0.5
RF6575	PA, DPDT, SPDT, LNA	2400 to 2500	28.0	20.0	3.3	160	3.5 x 3.5 x 0.5
RF6535	PA, DP2T, LNA	2400 to 2500	28.0	22.0	3.3	210	3.5 x 3.5 x 0.5
RF5745	PA, SP3T, LNA	2400 to 2500	30.0	23.0	3.3	230	3.0 x 3.0 x 0.5
RF6505	PA, SP2T, DP2T, LNA	2400 to 2500	28.0	27.0	3.6	500	3.5 x 3.5 x 0.6
RFFM6201	PA, SP2T, DP2T, LNA	2400 to 2500	25.0	23.0	2.0 to 3.6	175	5.0 x 5.0 x 1.0

